

GATE

GATE = General Architecture for Text Engineering

GATE is made up of three elements:

- ▶ An *architecture* describing how language processing systems are made up of components.
- ▶ A *framework* (or class library, or SDK), written in Java.
- ▶ A graphical *development environment* built on the framework.

GATE

CREOLE = GATE component model

- ▶ Pipeline of components
- ▶ Output of one components is input for another component

GATE

GATE comes with a large number of components

- ▶ Chunkers
- ▶ Taggers
- ▶ Stemmers
- ▶ Machine learning
- ▶ etc.

GATE interfaces to other technologies

- ▶ Databases
- ▶ Lucene
- ▶ Ontologies
- ▶ Web applications (Tomcat)
- ▶ etc.

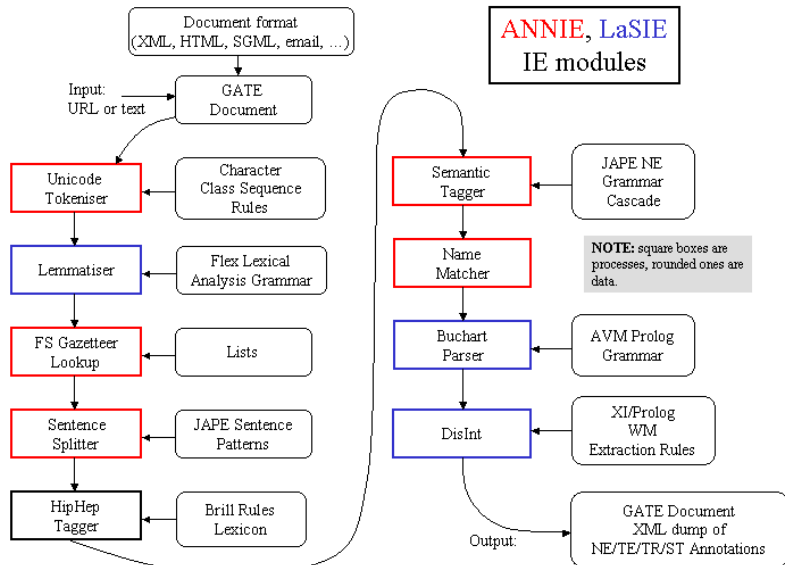
ANNIE

ANNIE = A Nearly-New Information Extraction System
(Precursor: LaSIE = Large Scale Information Extraction)

Information extraction system based on finite state algorithms and the JAPE language

Some of its components:

- ▶ Tokeniser
- ▶ Gazetteer
- ▶ Sentence Splitter
- ▶ Part of Speech Tagger
- ▶ Coreference resolution



ANNIE and LaSIE

GATE – our use of it

How we used GATE

Extraction of:

- ▶ Names
- ▶ Institutions
- ▶ Places

How we might use GATE

- ▶ Categorisation (genres, events)
- ▶ Clustering
- ▶ Finding similar documents

Improvements / problems

- ▶ Improvement of gazetteer lists for 19. century texts
 - ▶ Adaptation of existing gazetteer lists
 - ▶ Inclusion of additional gazetteer lists (Waterloo index)
- ▶ “Correct” bad OCR

Bad OCR 1

Leader, 23 October 1858, p. 1117

The proceedings in the case of **Arbuthnott v. Carden** were resumed on Monday, at Kingstown police-court. After a lengthened discussion, the magistrates decided on holding Mr. Carden to bail in the sum of 5000/. to keep the peace towards Miss **Arbuthnott**.

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Arbnthnott instead of **Arbuthnott**

Bad OCR 1

Northern Star, 31 May 1845, p. 3

Prodigious Ox.—One of the finest specimens of beef ever witnessed in Birmingham, was exhibited about a fortnight since at the back of Mr. Holder's, Rodney Inn, Coleshill-street. For neatness and symmetry of form, and fatness, the ox was the best of its species ever slaughtered in that town. The noble animal was of the Durham breed, by Mr. **Arbuthnot's** bull Despot; was bred by Mr. Baldwin, of Weston-on-Avon, near Stratford-on-Avon, and fed by Mr. Newbold, of Baginton, near Coventry. The gross weight was 2027 lb., or 253 st. 3 lb. A great portion of this extraordinary beast was sent to the metropolis.

Arbnthnot instead of **Arbuthnot**

Bad OCR 1

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Bad OCR 2

Saturday Analyst and Leader, 17 November 1860, p. 938

NEW THEATRE ROYAL ADELPHI.

Solo Proprietor and Manager, Mr. B. Webster.

Engagement for a limited number of nights of Miss Agnes Robertson and Mr. Dion Boucicault, who will appear every evening in **THE COLLEEN BAWN**.

On Monday and during the week

THE RIFLE BRIGADE.

Messrs. W. Smith, D. Fisher, C. Selby, Miss Woolgar, K. Kelly, and Mrs. Billington.

THE COLLEEN BAWN, Messrs. D. Boucicault, D. Fisher, Billington, Falconer, Stephenson, Romer, C. J. Smith, Miss Agnes Robertson, — Woolgar, Mrs. Billington, and Mrs. Chatterly; and

MUSIC HATH CHARMS.

Mr. D. Fisher and Miss K. Kelly. Commenced at seven.
Acting Manager Mr. W. Smith.

Blllinijton and BIIMngton instead of Billington

Akiioh Ilobbtrflon instead of Agnes Robertson

Bad OCR 2

Saturday Analyst and Leader, 17 November 1860, p. 938

NEW THEATRE ROYAL ADELPHI.

Solo Performer and Manager Mr. B. Webster.
Jingaymont for a limited number of nights of Miss
Aunos Itobortson and Mr. Dion' Boacloau, who will
appear every evening in The COLMEE* BAWN.

On Monday and during the
week-THE KING'S BIRTHDAY.

Moss. "W. Bmlth, D. Flalior, O. Solby, Miss Woplgar, K.
Kolly, and Mrs. Billinijton.

THE COTILDION BAWN", Mopar. P. Boiionult, D.
KIHhor, nil liny ton Falconer, stophoisou, Homer. C. J.
Smith, Miss Akiioh Ilobbtrflon, - Woolfrn, Mrs. BlIMngton.
and Miu Cliattorly; and
M. USIO HATH CHAKJM3.

Mr. P. Flwhor and JkHsa JC. Kolly. Commonoo 'nt sovon.
Acting Manager Mr. V. Bmlth.